Susovan Patra

Voice AI & ML Engineer
M.Tech CSE, IIT Kanpur
Email: susovanpatra00@gmail.com
GitHub: susovanpatra00 | LinkedIn: Susovan Patra | susovanpatra00.github.io

Summary

Voice AI and ML engineer with 2+ years of experience building real-time speech-to-speech systems, voice agent platforms and multilingual audio pipelines, as well as LLM and RAG systems for enterprise use. Brought a voice agent's end-to-end latency from about 2 s down to about 850 ms, analysed 6,000–10,000 calls a day in three languages, and generated about 20 hours of synthetic Indian-accent speech daily. Also write Triton kernels for LLM inference and training, and build computer vision tools.

Work Experience

- **Voice AI Engineer | Telephony and communications company** (Sept'25 - Present)
  - **Real-Time Voice Agent**
    - Built a real-time speech-to-speech voice agent and tested it under simulated concurrent load.
    - Reduced end-to-end latency from about 2 s to about 850 ms through model optimisation and pipeline parallelism on a single GPU, with no drop in response quality in benchmark evaluation.
  - **Post-Call Analytics System**
    - Built a multilingual post-call analytics system for 6,000–10,000 calls a day in Hindi, English and Tamil, with automatic language detection and about 90% transcription accuracy (internal evaluation).
    - Applied NLP and LLM-based analysis to extract customer pain points, intents and business-specific insights from transcripts.
  - **No-Code Voice Agent Platform**
    - Built a no-code platform for businesses to deploy phone voice agents through a visual flow builder, supporting greeting, data collection, API calls, call transfers and CRM integration.
    - Built a MongoDB-driven flow engine on Pipecat Flows that loads agent configuration at runtime per phone number, supporting multiple LLM, STT and TTS providers with no code changes for a new deployment.
    - Implemented the live voice pipeline over WebRTC at under 2 s end to end, with streaming STT and TTS, VAD and Indian accent support.
  - **Voice Data Generation & Training Pipeline**
    - Fine-tuned VibeVoice for Indian-accent speech and generated about 20 hours of synthetic audio a day, stored and managed with Hugging Face datasets and Cassandra.
    - Built a data loop that combines the synthetic audio with studio recordings to keep fine-tuning VibeVoice and a separate production TTS model.
  - **Real-Time Assistant for Call Centre Agents**
    - Built an assistant that takes call audio over WebSockets and uses sub-second LLM inference to give agents answers, FAQs and recommendations during live calls.
    - Built a low-latency RAG engine on Milvus and live pricing APIs for real-time inventory and price grounding, so agents no longer look things up by hand.
- **AI/ML Engineer | Motherson Technology Services Limited** (June'24 - Sept'25)
  - **Speech Diarization & Task Tracking System**
    - Built a system that splits meeting audio by speaker with pyannote and extracts tasks using keywords and RAG.
    - Added natural-language search over the segmented transcripts, so speaker-assigned tasks (for example, action words like 'send' or 'prepare') can be found with semantic search.
  - **Confidential Knowledge Chatbot for Motherson Group**
    - Built a secure RAG system over about 60 GB of confidential documents from 100+ group companies, answering queries in under 2.5 s.
    - Gave all the group companies one place to search, which cut down repeated work.
  - **M&A Contract Analysis**
    - Automated comparison of 4+ legal contracts of 200–300 pages each. An LLM extracts key clauses, summarises sections and flags differences, which cut due diligence review time by about a month.
    - Built a RAG chatbot so legal teams can ask questions about one or more contracts in natural language and get the relevant clauses and summaries without manual search.
  - **47×47 Multilingual Enterprise Translator**
    - Built and deployed a translation system for 47×47 language pairs using OpenAI and Helsinki-NLP Opus models. It handles .txt, .docx and .pdf files and keeps table structure through XML.

Projects

- **qwen2-triton-kernels: Fused Triton Kernels for Qwen2.5 Inference** (*Open source*)
  - Wrote fused Triton kernels for RMSNorm, SwiGLU and attention that patch into any Qwen2.5 model. Up to 26x faster in isolation.
  - About 1.1x faster end to end on the 7B and 14B models (A100, batch 8). No gain on the 0.5B model, where matmuls dominate.
- **qwen2-triton-training: LoRA Fine-Tuning of Qwen2.5-7B with Triton Kernels** (*Open source*)
  - Wrote Triton kernels for cross-entropy, RMSNorm and SwiGLU (forward and backward) for LoRA fine-tuning.
  - Cut peak VRAM by about 3.4 GB on an A100 (48.3 GB to 44.8 GB) at batch size 1, at a small throughput cost of about 4%.
- **SafeGuardAI: Computer Vision for Site Safety** (*Open source*)
  - Four modules: face recognition (InsightFace, FAISS, FastAPI), fire and smoke detection (YOLO11), Indian number plate reading (YOLO, PaddleOCR) and PPE compliance.
  - The PPE model (YOLO11s, trained on 17,507 images from three datasets) reaches 0.91 mAP50 on helmets and 0.93 on vests. Tested on public datasets only, not on live site cameras.
- **voice-pipeline-bench: Latency Benchmarks for ASR, LLM and TTS** (*Open source*)

- Benchmark suite for each stage of a voice pipeline: faster-whisper models, Qwen and Llama on vLLM (time to first token, tokens per second), and edge-tts and Kokoro (real-time factor).
- Found that faster-whisper small is the best speed and accuracy trade-off for a sub-second budget, since large-v3-turbo alone took about 626 ms on a single sentence.

- **Bolna Bhai: Modular Voice Agent for Indian Languages** (with Anand Therattil) (*Side project*)
  - Open-source ASR, LLM and TTS models, each behind its own WebSocket server and plugged into Pipecat, so a model can be swapped with a config change.
  - Covers AI4Bharat Indic Conformer, Vosk, Qwen, GPT-2, Indic Parler-TTS, Indri and Supertonic, with WebRTC transport.
- **MANN: Voice Agent Prototype for Mental-Health Triage** (with Anand Therattil) (*Side project*)
  - Prototype built on Bolna Bhai: Vosk for ASR, Phi-3-mini on vLLM and Supertonic for TTS, over WebRTC, with conversations saved after each call.
  - Urgency scoring and the dashboard are not built yet. Not a clinical tool.
- **Precise Position Using IMU Sensor Fusion and Machine Learning** (*M.Tech research, guide: Dr. Amitangshu Pal*)
  - Developed a sensor fusion algorithm combining accelerometer, compass and machine learning for GPS-free position estimation.
  - Reached 20–30 m accuracy over 4–6 km on real-world trajectory data. The algorithm works across sensors, including those in mobile phones.
- **Earlier projects:** text summarisation with fine-tuned Pegasus and Flask, wine quality prediction with MLflow, Docker and AWS EC2, news reporter with CrewAI.

SKILLS

- **Speech & Voice AI:** ASR, TTS, Speaker Diarization, Silero VAD, Pipecat, WebRTC, Deepgram, pyannote, Vosk, VibeVoice, Ultravox
- **LLMs:** vLLM, LoRA/QLoRA, Fine-Tuning, RAG, LangChain, LlamaIndex, Hugging Face, Inference Optimization
- **GPU & ML:** PyTorch, Triton, Transformers, Scikit-learn, MLflow
- **Computer Vision:** YOLO, InsightFace, PaddleOCR, OpenCV
- **Vector Search & Data:** Milvus, FAISS, ChromaDB, Pinecone, MongoDB, Cassandra
- **Backend & Cloud:** FastAPI, Flask, WebSockets, Apache Pulsar, Docker, AWS (EC2, S3, Lambda, ECS)
- **Programming Languages:** Python, C, C++, SQL

EDUCATION

Year	Degree	Institute
2022-2024	M.Tech (CSE)	Indian Institute of Technology Kanpur
2018-2022	B.Tech (CSBS)	Sister Nivedita University, Kolkata